

CHM1001 2025 term 1

Assignment 1

1. Multiple Choice Questions (Only one correct answer per question)

1. A photon has a wavelength of 250 nm. What is its energy?

- ☒ A. 7.96×10^{-19} J
- ☐ B. 5.12×10^{-19} J
- ☐ C. 3.24×10^{-18} J
- ☐ D. 2.48×10^{-19} J

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{250 \times 10^{-9}} = 7.96 \times 10^{-19}$$

2. Which transition in hydrogen results in the absorption of the shortest wavelength photon?

- ☒ A. $n=1 \rightarrow n=2$
- ☐ B. $n=2 \rightarrow n=4$
- ☐ C. $n=3 \rightarrow n=5$
- ☐ D. $n=4 \rightarrow n=6$

3. An electron in hydrogen is in $n=4$. How many different m_l values are possible?

- ☐ A. 4
- ☒ B. 7
- ☐ C. 9
- ☒ D. 16

$$l=0 \quad 1 \quad 2 \quad 3 \\ (+3+5+7)$$

4. The Heisenberg uncertainty principle means:

- ☐ A. Energy cannot be known
- ☐ B. Position is zero
- ☒ C. The more precisely momentum is known, the less precisely position is known

- D. Momentum and position can be measured exactly
5. Which electron configuration violates the Pauli exclusion principle?
- A. $1s^2 2s^2 2p^6 3s^2$
 - B. $1s^2 2s^2 2p^6 3s^2 3p^6$
 - C. $1s^2 2s^2 2p^5 3s^2 3p^5$
 - **D. $1s^2 2s^2 2p^7$**
6. In hydrogen, the energy depends only on:
- **A. n**
 - B. n and l
 - C. n and m_l
 - **D. n, l, m_l, m_s**
7. The first spectral line in the Balmer series results from:
- A. $n=2 \rightarrow n=3$
 - **B. $n=3 \rightarrow n=2$**
 - C. $n=4 \rightarrow n=2$
 - D. $n=5 \rightarrow n=2$
8. Maximum number of electrons for $n=5$ is:
- A. 25
 - **B. 32**
 - **C. 50**
 - D. 18
9. Which element ends with $4s^2 3d^5$?
- **A. Mn**
 - B. Fe
 - C. Cr
 - D. Co
10. Ground state for Cu is:
- A. $[Ar]4s^2 3d^9$
 - **B. $[Ar]4s^1 3d^{10}$**
 - C. $[Ar]4s^2 3d^{10}$

2
8
18
32
50
18
8

- D. $[\text{Ar}]4s^13d^9$

11. Which set of quantum numbers is not allowed?

- A. 2, 0, 0, $+\frac{1}{2}$
- B. 3, 2, -3, $-\frac{1}{2}$
- C. 4, 1, 0, $-\frac{1}{2}$
- D. 5, 3, 3, $+\frac{1}{2}$

12. The "3" in $3p^4$ represents:

- A. Protons
- B. Principal quantum number
- C. Angular momentum
- D. Magnetic quantum number

13. Greatest number of unpaired electrons in ground state:

- A. N 3
- B. Fe 4
- C. Cr 6
- D. O 2

14. Same configuration as Ar:

- A. Cl^-
- B. Ca^{2+}
- C. K^+
- D. All of these

15. Energy difference $n=2$ to $n=4$ in H atom:

- A. $2.18 \times 10^{-18} \text{ J}$
- B. $4.09 \times 10^{-19} \text{ J}$
- C. $1.23 \times 10^{-18} \text{ J}$
- D. $3.40 \times 10^{-19} \text{ J}$

$$\Delta E = -(6.63 \times 10^{-34}) (3 \times 10^8) (1.097 \times 10^7) \left(\frac{1}{4^2} - \frac{1}{2^2} \right)$$

$$\approx 4.091 \times 10^{-19}$$

16. Which statement best describes the modern periodic law?

- A. Properties of elements repeat periodically with atomic mass.
- B. Properties of elements repeat periodically with atomic number.

- C. Properties vary randomly with protons.
- D. Properties depend only on number of neutrons.

17. Which element is a metalloid?

- A. Al
- B. Si
- C. S
- D. P

18. Effective nuclear charge is:

- A. The total charge of all electrons in an atom.
- B. The net positive charge experienced by a specific electron.
- C. Always equal to the number of protons.
- D. Independent of electron shielding.

19. Across a period from left to right, Z_{eff} generally:

- A. Increases
- B. Decreases
- C. Stays constant
- D. Doubles

20. Atomic radius trend across a period:

- A. Increases
- B. Decreases
- C. Stays the same
- D. First increases, then decreases

21. Compared to their neutral atoms, cations:

- A. Are larger due to loss of protons.
- B. Are larger due to increased shielding.
- C. Are smaller due to decreased electron–electron repulsion.
- D. Have the same radius.

22. In an isoelectronic series, radius decreases with:

- ☒ A. Increasing nuclear charge.
- ☐ B. Increasing number of electrons.
- ☐ C. Increasing shielding.
- ☐ D. Decreasing Z_{eff} .

23. The first ionization energy is:

- ☐ A. Energy released upon adding an electron.
- ☒ B. Energy required to remove the outermost electron from a gaseous atom.
- ☐ C. Energy needed to remove a core electron.
- ☐ D. Energy change for nuclear processes.

24. First ionization energy across a period:

- ☒ A. Increases
- ☐ B. Decreases
- ☐ C. Remains constant
- ☐ D. Randomly fluctuates

25. Electron affinity is:

- ☐ A. Energy to remove an electron from a gaseous atom.
- ☒ B. Energy change when an electron is added to a gaseous atom.
- ☐ C. Always negative.
- ☐ D. Independent of nuclear charge.

26. Among the following, which has the most negative electron affinity value?

- ☐ A. Na
- ☐ B. O
- ☒ C. Cl
- ☐ D. Ar

27. Metal oxides generally produce solutions that are:

- A. Acidic
- **B. Basic**
- C. Neutral
- D. Amphoteric only

28. Which alkali metal is most reactive with water?

- A. Li
- B. Na
- C. K
- **D. Cs**

29. Halogens typically exist as:

- A. Monoatomic gases
- **B. Diatomic nonmetal molecules**
- C. Noble gases
- D. Metals

30. Noble gases are:

- A. Reactive diatomic gases
- **B. Unreactive monoatomic gases**
- C. Solid metals at room temperature
- D. Reactive with acids

2. Short-Answer/Calculation Questions

- Calculate the frequency of light with $\lambda = 600 \text{ nm}$. $f = \frac{c}{\lambda} = \frac{3 \times 10^8}{6 \times 10^{-7}} = 5 \times 10^{14}$
- Calculate the wavelength (λ) for an electron moving at $1.5 \times 10^6 \text{ m/s}$. $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \cdot 1.5 \times 10^6} = 4.85 \times 10^{-10} \text{ m} = 0.485 \text{ nm}$
- What is the maximum number of electrons in the $n=4$ shell? **32 electrons**
- Write the electron configuration for Zn^{2+} . $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$
- Calculate the wavelength (λ) for the $n=3 \rightarrow n=1$ transition in a hydrogen atom. $\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = (1.097 \times 10^7) \left(\frac{1}{1^2} - \frac{1}{3^2} \right) \rightarrow \lambda = 103 \text{ nm}$
- Predict which species has the largest atomic radius: S^{2-} , Cl , K^+ , Ca^{2+} . **S^{2-} has largest atomic radius (least attractive force from the nucleus)**
- Why does the electron affinity of noble gases appear positive? **p-sublevel of noble gases are full, thus adding another electron to them would make them more unstable. Hence, it is considered unfavorable, or require energy to do so.**

(unfavorable).